NAME ____________________________________

ADM NO ________________

DATE ________________

231/1
BIOLOGY
(THEORY)
PAPER 1

TIME: 2 HOURS

END OF TERM TWO FORM FOUR EXAMINATION, 2019

INSTRUCTIONS TO CANDIDATES

1. Write your name and admission number in the spaces provided above.
2. Write the date in the spaces provided above.
3. Answer **ALL** the questions in the spaces provided.
4. Additional pages **MUST** not be inserted.
5. Candidates may be penalized for false information and even wrong spellings of technical terms.
6. This paper consists of **13** printed pages.
7. Candidates should check to ensure that all pages are printed as indicated and no questions are missing.

FOR OFFICIAL USE ONLY

QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORE
1 – 30	80	

1. Differentiate between locomotion and movement. (2mks)

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…………………………………………………………………………………………………

2. (a) Identify the organelle that is likely to be most abundant in the phagocytes.

Explain. (2mks)

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…………………………………………………………………………………………………
…………………………………………………………………………………………………

(b) State the function of:

(i) Grana of chloroplasts. (1mk)

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(ii) Golgi vesicles. (1mk)

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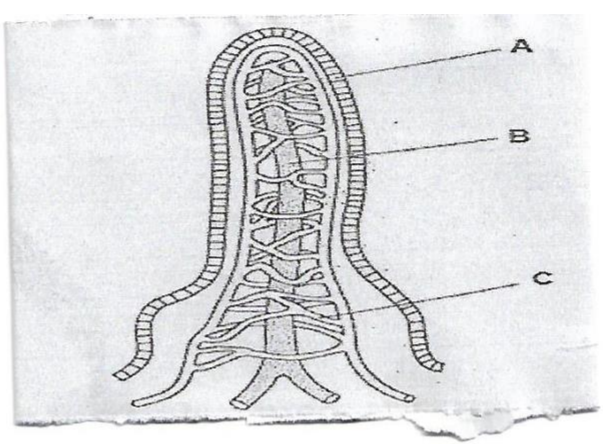
3. The diagram below represents a villus of the human alimentary canal.



(a) Name the substances that are absorbed through the structures labelled B and C. (2mks)

B…………………………………………………………………………………………

C…………………………………………………………………………………………

(b) State one adaptation of the carnassial teeth in carnivores. (1mk)

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4. Explain why the lens and the mirror should not be touched with fingers. (2mks)

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5. Explain what happens when a marine amoeba is transferred to distilled water. (3mks)

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6. Name the cell structure responsible for:

(i) Turgor pressure (1mk)

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(ii) Wall pressure (1mk)

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7. (a) State one similarity and one difference between epithelial and epidermal tissues.

Similarity (1mk)

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Difference (1mk)

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(b) State two importance of cell membrane having electric charges. (2mks)

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8. Explain the significance of diffusion to plant pollination. (1mk)

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9 Explain the process of starch digestion in the duodenum. (3mks)

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10. Glucose and oxygen gas are produced as a result of photosynthesis.

(i) Where do the oxygen atoms in glucose come from? (1mk)

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(ii)Where do the atoms in the oxygen gas come from? (1mk)

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11. At temperatures above 40oc the rate of transpiration falls. Explain. (1mk)

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12 Give the significance of the endothelium in arteries. (1mk)

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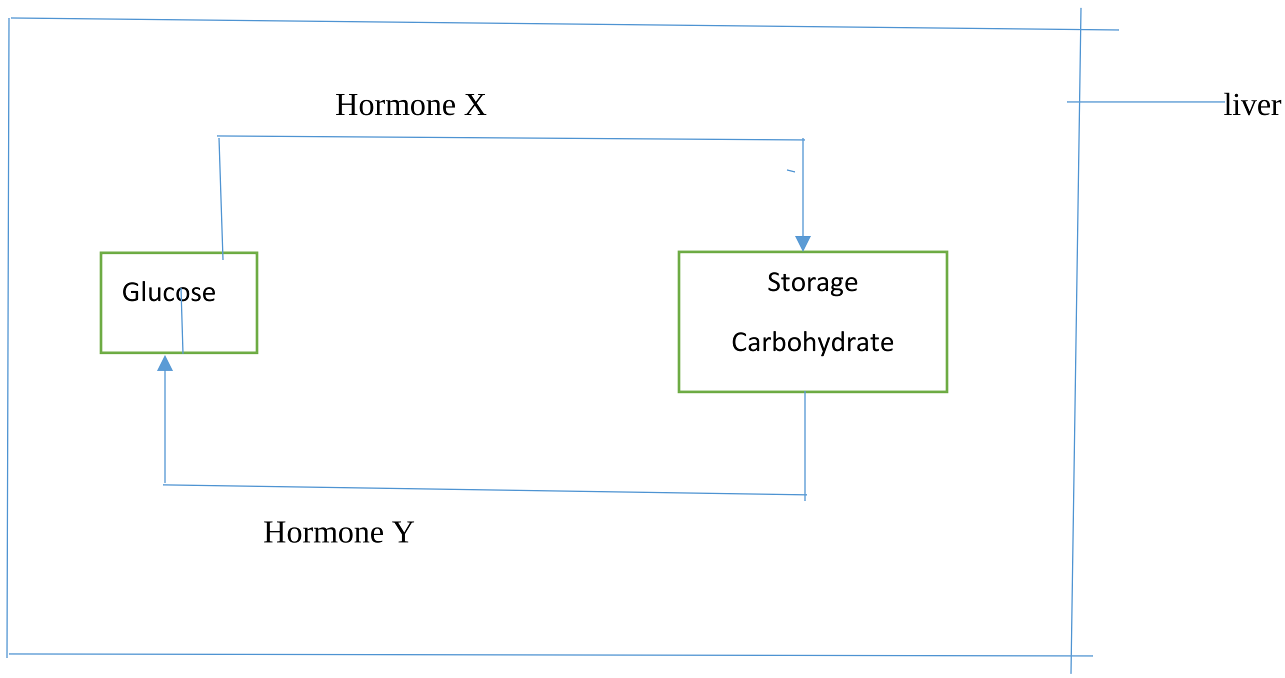
13. The homeostatic control of blood glucose concentration carried out by the human liver is shown on the diagram.



(a) Name the storage carbohydrate found in the liver. (1mk)

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…………………………………………………………………………………………

(b) (i) Name the hormone X and Y (2mks)

Hormone X…………………………………………………………………………

Hormone Y…………………………………………………………………………

(b) Name the organ that produces hormones X and Y (1mk)

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14. Differentiate between lenticels and spiracles. (1mk)

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15 Explain the events that follow oxygen debt. (3mks)

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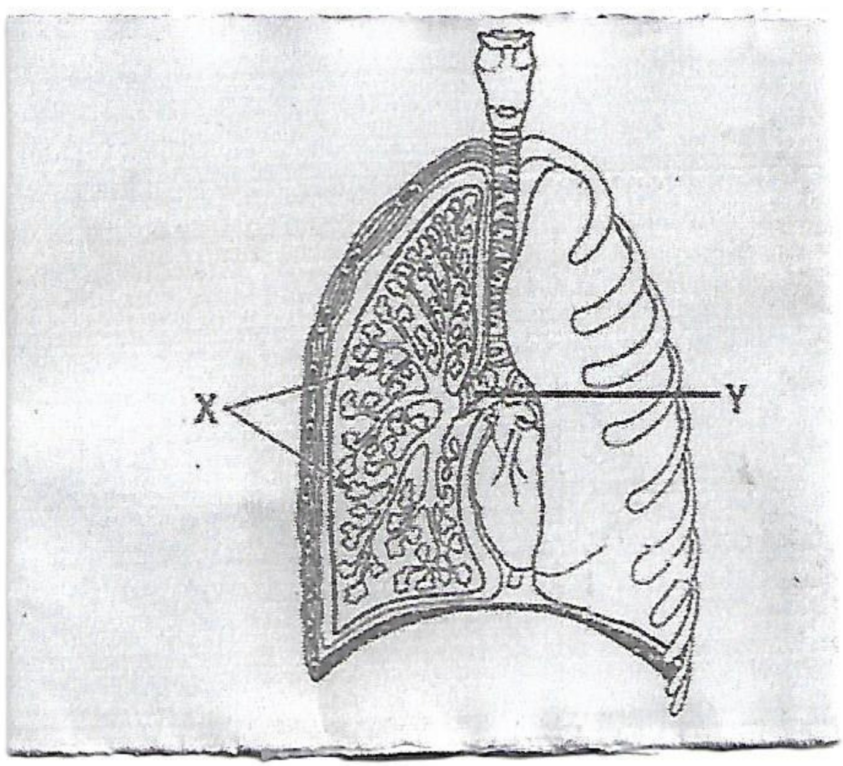
16. The diagram below illustrates part of the human respiratory system.



(a) Name the organism that may cause infection to the parts marked X. (1mk)

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(b) State one way in which part Y is suited to its function. (1mk)

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17. State the functional difference between prostate glands and cowpers' gland. (2mks)

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18. State two methods by which fossils were formed. (2 marks)

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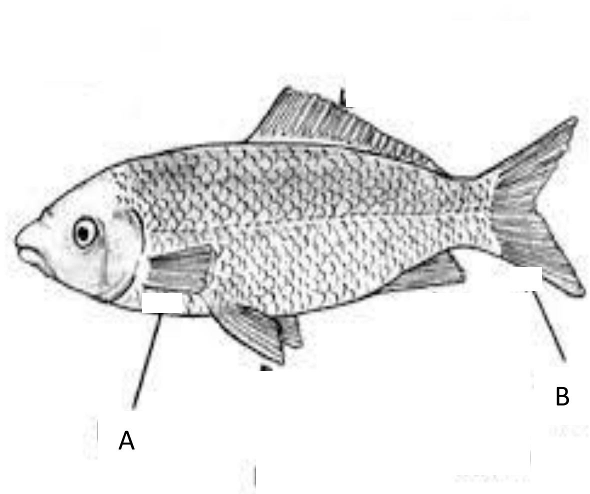
19. The diagram below shows the external structure of Tilapia.



(a) A part from swimming, state other functions of the fins labelled A & B (2mks)

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(b) State two functions of swim bladder in a fish. (2mks)

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20. (i) State two importance of DNA molecule. (2mks)

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(ii) What is DNA replication? (1mk)

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(iii) A DNA strand sequence had the following base sequence TAC GCT. What is the sequence of M- RNA strand copied from this DNA portion? (1mk)

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21. Differentiate between aerotaxis and rheotaxis (2 marks)

Aerotaxis

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Rheotaxis

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…………………………………………………………………………………………………

22. Which structure in the ear detects

(a) Sound waves. (1mk)

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(a) Change in posture. (1mk)

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23. (a) What do similarity of wings of bats and those of insects illustrate? (1mrk)

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(b) Name three evidences that show that organic evolution has taken place. (3 mrks)

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24. Explain what is meant by a test-cross as used in genetics. (1mrk)

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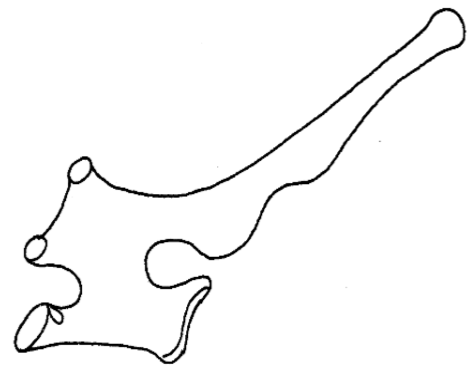
(a) Determine the probability of a couple with blood group AB getting a child with blood group **B**. (Show your working). (3mks)

25. The diagram below represents a mammalian vertebra.



(a) Identify the vertebra represented above. (1mk)

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(b) Give a reason for your answer. (1mk)

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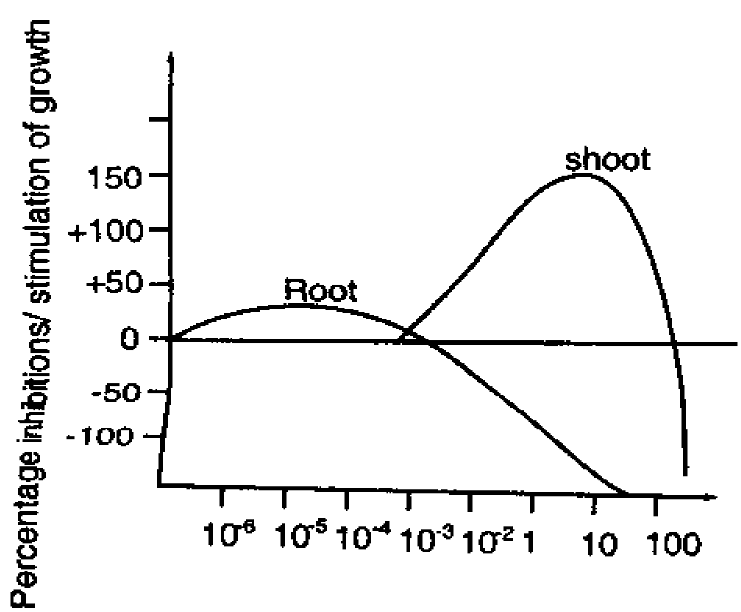
26. Below is a graphical representation of the effects of different concentration of auxins on shoot and root growth. Study it carefully and then answer the questions that follow.



Auxin concentration (ppm

(a) Identify **any two** conclusions that can be drawn from the graph. (2mks)

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(b) Name the growth hormone responsible for ripening of fruits. (1mk)

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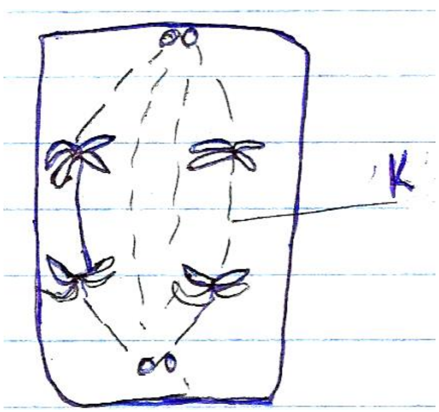
27. The diagram below represents a stage during cell division.



i) Identify the stage of cell division. (1 mrk)

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ii) Give two reasons for your answer (a) above. (2 mrks)

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iii) Name the structures labeled K. (1 mrk)

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28. Name the causative agent for the following diseases;

a) Typhoid (1 mrk)

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b) Syphilis (1 mrk)

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29 (a).Name **three** supportive tissues in plants. (3mks)

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(b)Name the type of muscles found in the gut. (1mk)

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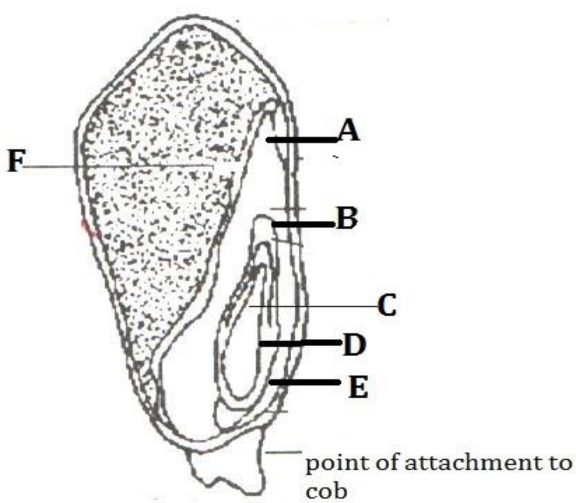
30. The diagram below shows the structure of a monocotyledonous seed



(a) Name the parts B and D (2mrks)

B………………………………………………………………………………………

D………………………………………………………………………………………

(a) Name the parts that would stain blue black with iodine solution. (2mrks)

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